

Takahiro Matsunaga

Curriculum Vitae

Personal Information

Name: Takahiro Matsunaga

Affiliation: Department of Civil Engineering, the University of Tokyo

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Education

Ph.D. in Engineering

Since Oct, 2025

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

M.S. in Engineering (Outstanding Paper Award)

Apr, 2024 – Sep, 2025

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

B.S. in Engineering (Best Paper Award)

Apr, 2020 – Mar, 2024

Behavior in Networks Studies Unit (Supervisor: Eiji Hato)

Department of Civil Engineering, the University of Tokyo

Peer-Reviewed International Journal Papers

- [1] Takahiro Matsunaga and Eiji Hato. “Mean field dynamics for pedestrian interaction in quantized observation-based 3D route choice modeling”. In: *Transportation Research Part C: Emerging Technologies* 186 (2026), p. 105608. ISSN: 0968-090X. DOI: <https://doi.org/10.1016/j.trc.2026.105608>. URL: <https://www.sciencedirect.com/science/article/pii/S0968090X26000963>.

Peer-Reviewed National Journal Papers (in Japanese)

- [1] 松永 隆宏, 山本 航介, and 羽藤 英二. “累積参加型抽選インセンティブに基づく混雑制御の動学的性質と実証分析”. In: *都市計画論文集* 61.3 (2026).
- [2] 松永隆宏 and 羽藤英二. “言語生成モデルに基づく首都圏移動活動経路の予測”. In: *都市計画論文集* 60.3 (2025), pp. 1527–1534.
- [3] 松永 隆宏 and 羽藤 英二. “BLE 観測の不確実性を考慮した屋内 3 次元経路選択モデルの推定”. In: *都市計画論文集* 59.3 (2024), pp. 1683–1690.

Peer-Reviewed International Conferences

- [1] Takahiro Matsunaga and Eiji Hato. “Dynamic Congestion Control in Metropolitan Railways using Generative Language Models: Approach by Bid-free Reverse Auctions”. In: *Transportation Research Board Annual Meeting (TRB 2025)*. Peer-reviewed. Washington, D.C., USA, 2026.
- [2] Takahiro Matsunaga and Eiji Hato. “Partial Controllability in Mean Field Incentive Dynamics”. In: *European Association for Research in Transportation (hEART 2026)*. Peer-reviewed. Paris., France, 2026.
- [3] Takahiro Matsunaga and Eiji Hato. “Activity Generation using Panel Data of Day-to-day Activity-travel Pattern in Tokyo Metropolitan Area”. In: *The 29th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2025)*. Peer-reviewed, oral presentation. Hong Kong, 2025.
- [4] Takahiro Matsunaga and Eiji Hato. “Interaction of 3D pedestrian flow in a congested railway station: Structural estimation based on Mean Field Game theory”. In: *The 12th International Conference on Pedestrian and Evacuation Dynamics (PED 2025)*. Peer-reviewed, accepted, oral presentation. Prague, Czech Republic, 2025.
- [5] Takahiro Matsunaga and Eiji Hato. “Joint Estimation of Latent OD Matrix and 3D Route Choice Model Across Railway Station Boundaries based on Manifold Learning”. In: *The 28th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2024)*. Peer-reviewed, oral presentation. Hong Kong, 2024.
- [6] Takahiro Matsunaga and Eiji Hato. “Three-Dimensional Pedestrian Route Choice Model with Quantization of Observational Uncertainty”. In: *IEEE International Conference on Intelligent Transportation Systems (ITSC)*. Peer-reviewed, oral presentation. Edmonton, Canada, 2024.
- [7] Takahiro Matsunaga and Eiji Hato. “Three-Dimensional Pedestrian Route Choice Model with Quantization of Observational Uncertainty”. In: *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE. 2024, pp. 1–8.

Oral Presentation without Review (in Japanese)

- [1] 山本 航介, 松永 隆宏, and 羽藤 英二. “ポイントインセンティブによる駅構内経路選択変化に関する実証分析—新宿駅乗換改札誘導実験を対象として—”. In: **土木計画学研究発表会・秋大会**. 2026.
- [2] 松永 隆宏 and 羽藤 英二. “部分観測型交通需要の平均場制御—Causal Transformer による動学的選好の導入—”. In: **土木計画学研究発表会・春大会**. 2026.
- [3] Takahiro Matsunaga and Eiji Hato. “Incentive Design for Congestion Control of Public Transport Demands based on Reverse Auction”. In: **土木計画学研究発表会・秋大会**. 2025.
- [4] 松永 隆宏 and 羽藤 英二. “複雑な day-to-day 活動動態学習に基づく首都圏アクティビティ生成モデル”. In: **交通工学研究発表会**. 2025.
- [5] 松永 隆宏 and 羽藤 英二. “駅構内における三次元経路選択モデルの多様体推定”. In: **土木学会全国大会**. 2024.
- [6] 鈴木 健史, 齋藤 功次, 山崎 淳, 本田 聖一朗, 羽藤 英二, 石井 良治, and 松永 隆宏. “渋谷駅山手線線路切換工事時における多様な観測データを用いた行動変容の実態分析”. In: **土木計画学研究発表会・秋大会 (2023)**.

Awards

- [1] *Outstanding Master Thesis*. The Department of Civil Engineering, The University of Tokyo, 2025.
- [2] *Outstanding Research Presentation*. Japan Society of Traffic Engineers, 2025.

- [3] *Outstanding Student Paper Award (2nd Place)*. Hong Kong Society for Transportation Studies (HKSTS), 2025.
- [4] *Best Graduation Thesis (Dean's Award)*. The Department of Civil Engineering, The University of Tokyo, 2023.

Fellowships

- [1] *JSPS Research Fellow (DC1)*. Japan Society for the Promotion of Science, Apr, 2026 -.
- [2] *SPRING GX (Support for doctoral student)*. The University of Tokyo, Fostering Advanced Human Resources to Green Transformation, Oct, 2025 -.
- [3] *WINGS-GLAFS (Support for master and doctoral student)*. The University of Tokyo, Institute of Gerontology, Oct, 2024 - Sep, 2025.

Other Public Outputs (in Japanese)

- [1] 羽藤 英二, 小川 大智, 古橋 郁一, 松永 隆宏, 今村 啓太, and 須賀 理帆. **現代行動理論の基礎と応用**. Behavior in Networks Studies Unit, The University of Tokyo. 2026. URL: <https://github.com/bin-utokyo/stone-ship>.